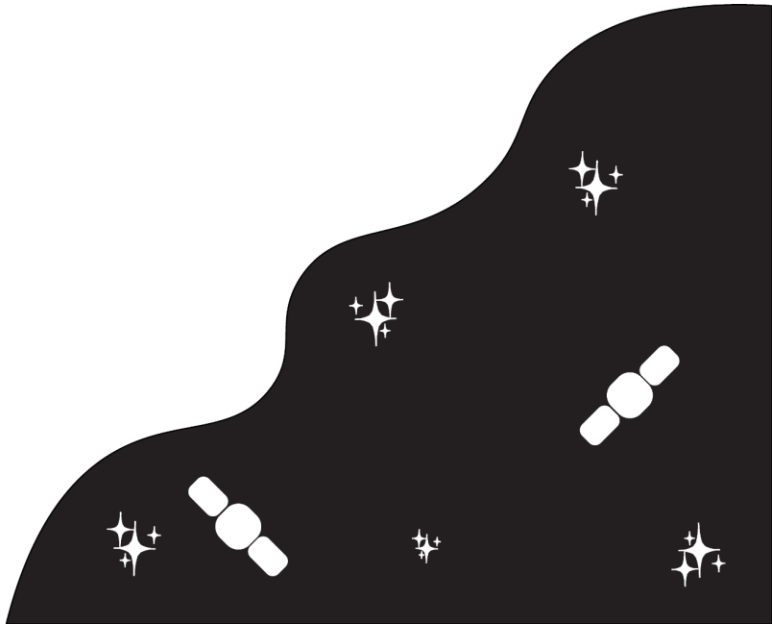
A Review of Early Excess Flux SNe Ia and Their Hosts

Riley Webster and Anais Möller

Honours Project A

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Rubin and LSST

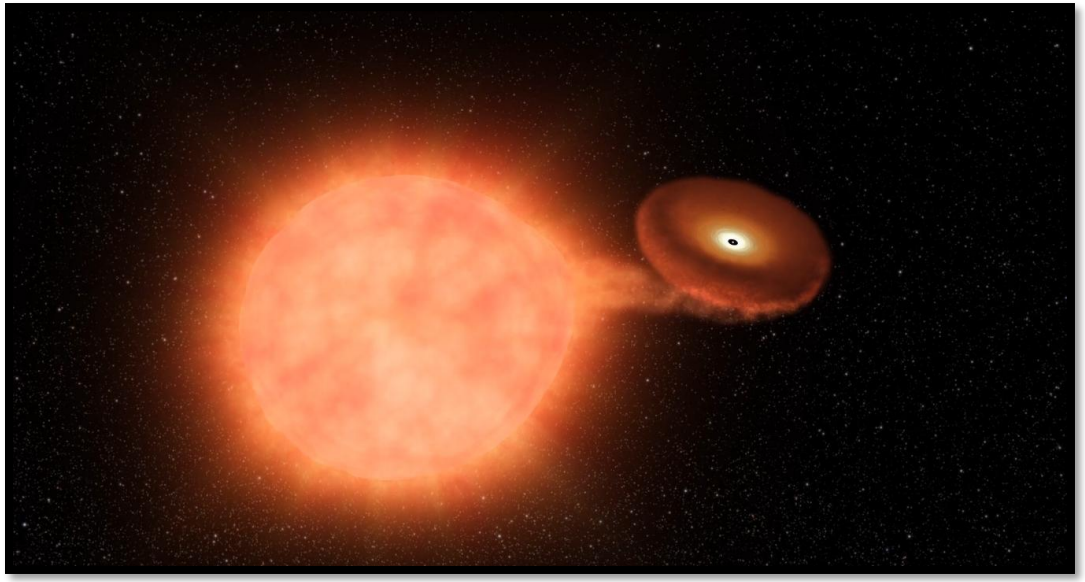
- Rubin telescope
  - 8.36 metre aperture
  - 320.5 – 1084.5 nm
- LSST
  - Observe the southern skies
  - Produce 10 million alerts per night



(*Vera C. Rubin Press Release*)

What are SN Ia?

- The explosion of a carbon-oxygen white dwarf

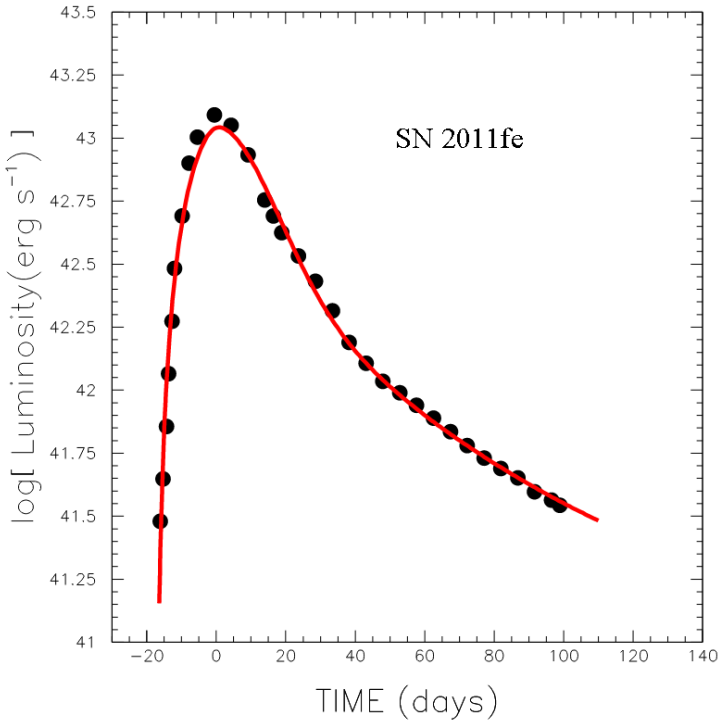




(*NASA Image and Video Library*)

SNe Ia – Why?

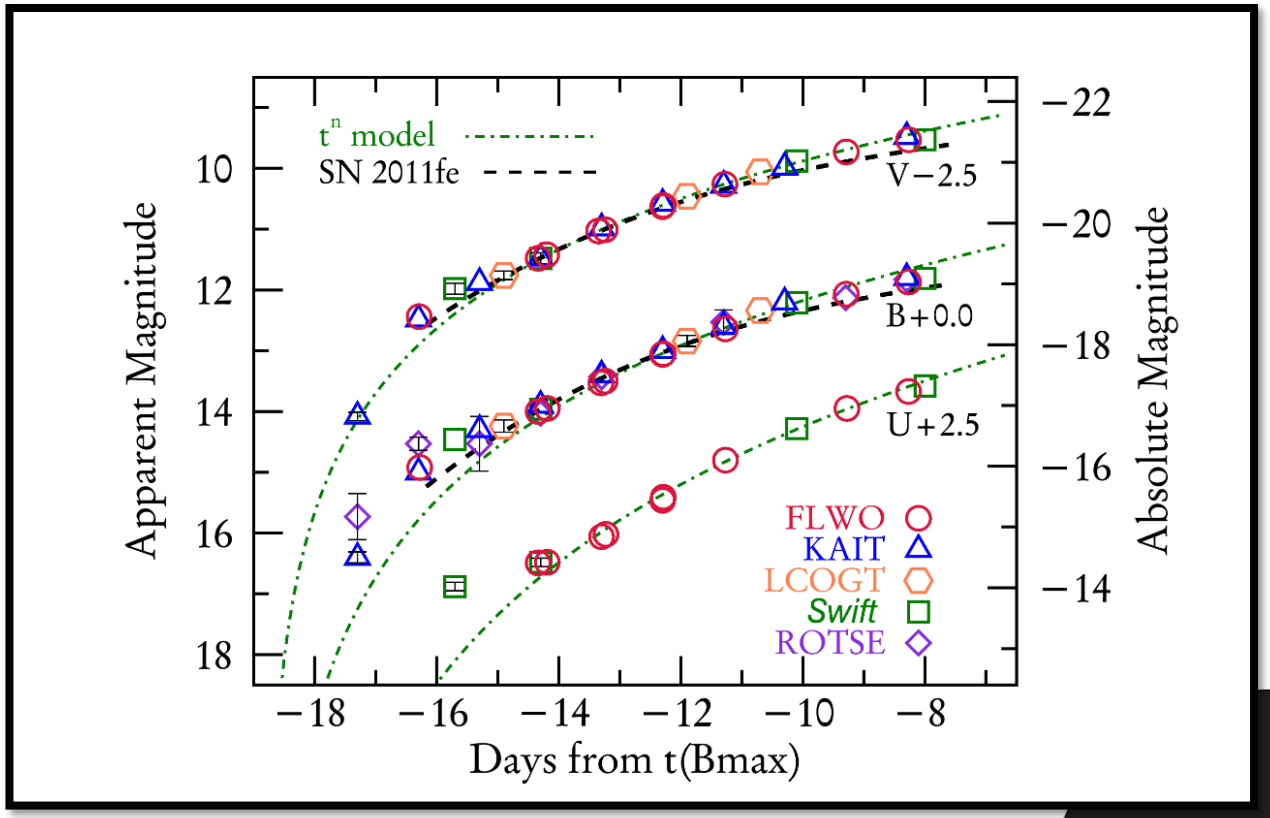
- SNe Ia are standardizable candles



(Dado & Dar, 2015.)

Early Excess Flux

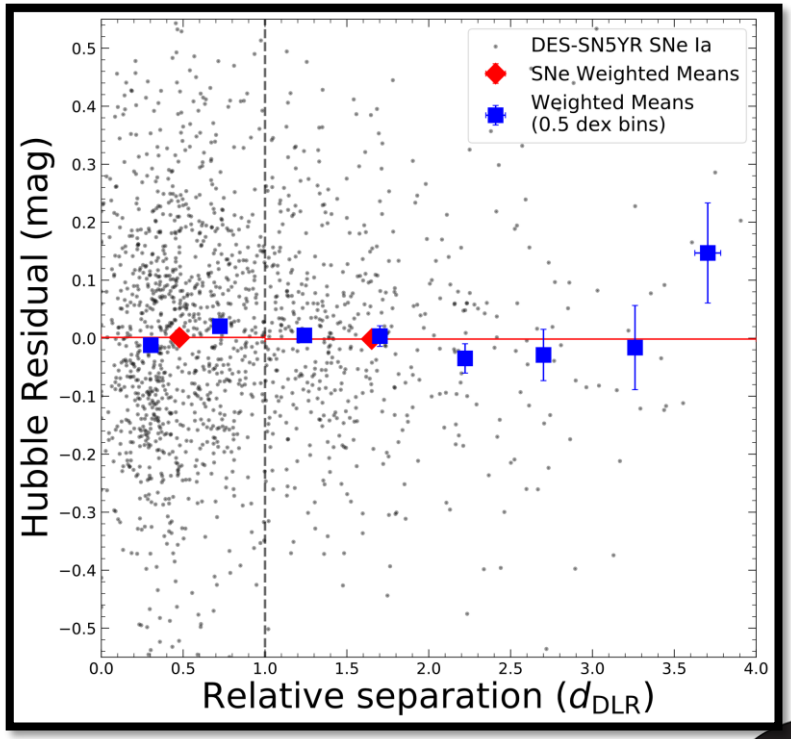
- In some cases, observation of SNe Ia are early and produce a bump in the light curve.
- Case studies have proposed companion interaction, clumping of nickel 56 and interactions with CSM



(Marion et al., 2016)

Host Galaxy Environments

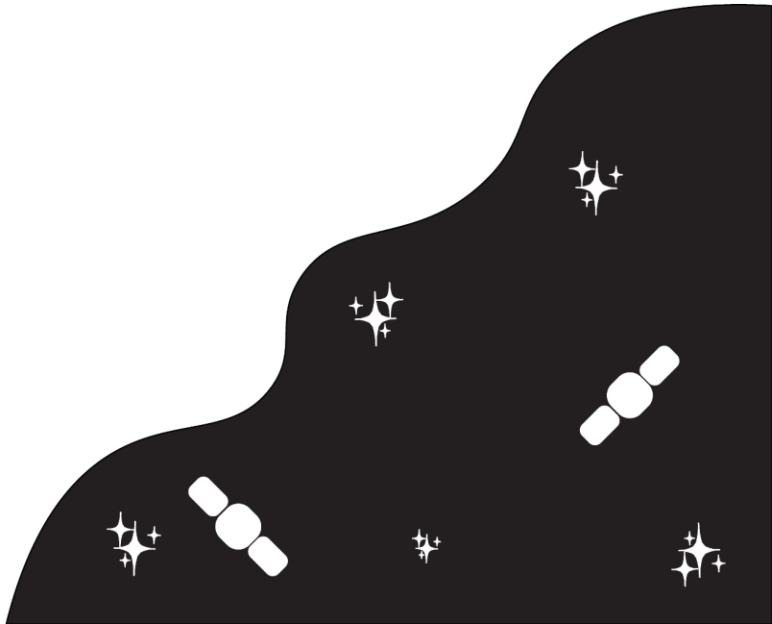
- To improve standardisation, we search for correlations with Host galaxy properties.



(Toy et al., 2025)

Finding SNe Ia

- Use Fink SuperNNova and earlySNIa classifiers to identify SNe Ia candidates.
  - Through cross matching with galaxy catalogues
  - Object lacks a history of variability



Host Galaxy Properties

- Use fits of the host galaxy fluxes in the various bands of the Vera C. Rubin telescope to determine the following properties.
  - SFR
  - Stellar Mass
  - Colour
- Use the position of the SNe Ia to determine the DLR.



What's Next

- Use SALT modelled light curves to find flux residuals in the SNe Ia.
  - This may give us a sample of early excess flux SNe Ia.
- If sample size is significant, the correlations with the host galaxy properties of the early excess flux will be found!

